

TACO-Gold

From Trump Retreat Risk to Gold Forecasting

Group 5

Project Presentation

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Introduction

Gold Price and Selected Trump Retreat Episodes



Trump's pressure-then-soften episodes often coincide with visible gold repricing. This motivates our core question: can we turn that pattern into a useful forecasting signal?

Research Questions

TACO: *Trump Always Chickens Out*. It refers to Trump's tendency to back off after pressure or tariff threats.

Question 1

Can Trump's "pressure-then-soften" text pattern be transformed into a tradable signal?

Question 2

Can that signal improve gold-return forecasting beyond standard macro and market information?

Working Pipeline

Text pattern $\rightarrow$ event-level probability $P_taco \rightarrow$ gold return forecasting

What Exactly Are We Modeling?

Not the broad popular claim

We are **not** trying to model the vague statement that Trump “often TACO” in a general political sense.

Operational definition in this project

- Start from a pressure-like anchor text
- Observe Trump’s own follow-up public texts within the next **48 hours**
- Detect whether a retreat signal appears: pause, delay, exemption, narrowing, or a shift toward negotiation

Output: event-level probability P_{taco}

Data and Labeling

Data Sources

Data block	What we collect	Primary source
Trump first-term text	Tweets, timestamps, raw text	Kaggle dataset
Trump second-term text	Truth Social posts, timestamps, raw text	Public GitHub archive
Market stress	SPX drawdown, VIX change, related controls	FRED
Gold target series	Gold daily close and return series	Alpha Vantage
Attention proxy	Tariff / war / inflation search attention	Google Trends

Note: The second-term Truth Social archive comes from the public GitHub project *stiles/trump-truth-social-archive*.

Sample Split and Labeling

Training and Test Windows

- First-term training window:
2017-01-20 to 2021-01-08
- Second-term training window:
2025-01-20 to 2025-07-31
- Test set / primary holdout:
2025-08-01 to 2025-10-25

Biden-period observations and the platform gap are excluded from the main training and test design.

Stage 1 Setting

- Event-level dataset constructed from policy threats
- **Severely imbalanced** (rare retreat events)
- Focus on **probability estimation** rather than hard classification

Labeling

Labels are generated via a **rule + DeepSeek pipeline**, identifying whether a directional policy retreat occurs within a 48-hour window.

Why Stage 1 Is a Rare-Event Problem

Typical balanced classification

- Positive and negative classes are closer in size
- Accuracy and ROC-style metrics are often informative
- The goal is usually overall classification correctness

Our Stage 1 task

- Positive events are very rare ($< 5\%$)
- We care about ranking the few important positives near the top
- **PR-AUC** is more useful than plain accuracy here

Intuition: PR-AUC focuses on whether the model's high-risk predictions are truly positive and whether rare positive cases are captured early.

How We Handle the Rare-Event Setting

- **Reweighting:** increase the importance of rare positives during training.
- **Selection metric:** compare models mainly with **PR-AUC**.
- **Calibration:** make the final *P_taco* usable as a probability signal.

Model-level handling

- Linear models: class reweighting to improve sensitivity to rare positives
- Tree models: loss weighting based on the positive/negative class ratio
- Final comparison: probability quality rather than only hard-label accuracy

References: King and Zeng (2001); Saito and Rehmsmeier (2015).

Method

Validation Design and Leakage Control

- **Nested CV:** model tuning and model comparison are separated.
- **OOF probabilities:** Stage 2 only consumes Stage 1 out-of-fold or true holdout probabilities.
- **Primary holdout:** *2025-08-01* to *2025-10-25* is reserved for final out-of-sample reporting.
- **Purged CV + embargo:** used to reduce neighboring-sample contamination and boundary leakage.
- **Lagged controls:** macro and market variables are lagged so only information available before the target day enters prediction.

Key rule

Stage 2 never uses in-sample Stage 1 probabilities.

Why a Two-Stage Design?

One-stage

Raw text sentiment aggregation



Gold return prediction

Two-stage

Text + pressure features



Event-level P_taco



Gold return prediction

Stage 1 acts as a **supervised signal compression** step: it turns noisy text into a task-aligned intermediate probability signal.

What Goes into Stage 1?

Feature family	Examples	Role
Text sentiment	FinBERT <i>pos</i> / <i>neu</i> / <i>neg</i>	Capture the semantic tone of the anchor text
Candidate intensity	Keyword score, candidate priority, follow-up density	Describe whether the anchor looks like a strong pressure event
Event metadata	Source, timing, term identifier	Preserve contextual event information
Pressure features	SPX stress, VIX-related pressure signals	Capture market conditions around the event
Attention proxy	Tariff / war / inflation attention	Optional public-attention signal

FinBERT provides **positive / neutral / negative** probabilities for each anchor text. These are Stage 1 input features, not the final gold-forecast output.

Stage 1 Result: Can We Learn a Usable P_taco?

Rank	Model	Prob.	PR-AUC	Brier	ECE
1	linear_svm	sigmoid	0.0837	0.0442	0.0218
2	linear_svm	isotonic	0.0679	0.0448	0.0302
3	xgboost_classifier	sigmoid	0.0464	0.0448	0.0250
4	xgboost_classifier	isotonic	0.0649	0.0455	0.0278
5	logistic_regression	isotonic	0.0735	0.0456	0.0309
6	logistic_regression	sigmoid	0.0669	0.0461	0.0280
7	xgboost_classifier	raw	0.0621	0.0531	0.0534
8	logistic_regression	raw	0.0856	0.2601	0.4520

Best probability variant under the outer-CV selection rule: **linear_svm + sigmoid**

Note: although *logistic_regression raw* has a high PR-AUC, its probability quality is much worse in Brier and ECE, so it is not selected.

What Goes into Stage 2?

Feature family	Representative inputs
Daily P_taco aggregates	p_taco_any , p_taco_top1 , $p_taco_tail_010$, $p_taco_any_decay_2d$
Lagged macro / market controls	DXY return, real yield change, VIX, SPX stress
Cross-asset structure	Gold-SPX rolling correlation

Target definition

$$\text{Gold_Return}_e = \log \left(\frac{\text{GoldClose}_{\tau(e)}}{\text{GoldClose}_{a(e)}} \right)$$

- $\tau(e)$: first valid trading day after the event
- $a(e)$: nearest valid trading day before the event

Stage 2 Final Model Selection

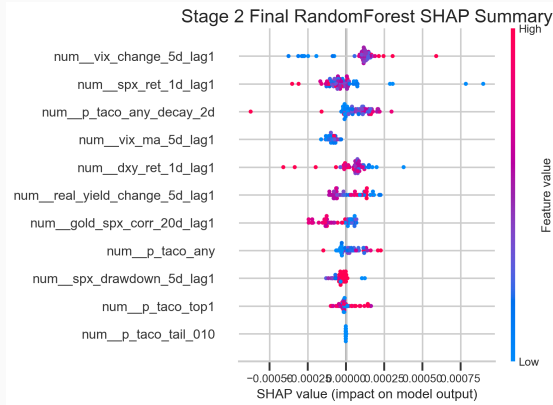
Rank	Model	RMSE	MAE
1	RF	0.0098	0.0072
2	Ridge	0.0099	0.0072
3	XGB	0.0100	0.0074

Final selection is based on **outer-CV RMSE and MAE**; the holdout is used only for final out-of-sample reporting.

Selected model: *svm_sigmoid + RF + two_stage_v2_core*

Holdout RMSE comparison: RF = 0.0153, Ridge = 0.0156, XGB = 0.0153.

Interpretation: What Does the Final Model Use?



P_taco-related features enter the important-feature layer together with lagged macro-financial controls, suggesting that the signal is not merely duplicating standard market information.

Results

Main Result I: Does the Signal Add Value Beyond Standard Baselines?

Benchmark	RMSE	MAE	R^2	Sign Acc.
Random Walk	0.015593	0.009891	-0.061624	0.000000
Macro-only Linear	0.015593	0.009773	-0.061553	0.590164
Macro-only Tree	0.015432	0.009682	-0.039723	0.688525
Two-stage Main Model	0.015349	0.009605	-0.028609	0.672131

This slide answers the first research question. The two-stage main model improves on random walk and linear macro baselines, and delivers the best overall RMSE among the benchmark set.

Holdout RMSE: two-stage = 0.015349 vs macro-only comparator = 0.015409.

Main Result II: Is Two-Stage Better Than Direct Text-to-Return Modeling?

Model	RMSE	MAE	R^2	Sign Acc.
Macro-only	0.015409	0.009675	-0.036685	0.622951
Single-stage	0.015417	0.009725	-0.037789	0.639344
Two-stage Main Model	0.015349	0.009605	-0.028609	0.672131

This slide answers the second research question. In the final comparison, the ordering is:

direct text < macro-only < two-stage

Why Is the Out-of-Sample R^2 Negative?

Out-of-sample R^2 formula

$$R^2_{\text{oos}} = 1 - \frac{\sum_{t \in \text{test}} (y_t - \hat{y}_t)^2}{\sum_{t \in \text{test}} (y_t - \bar{y}_{\text{test}})^2}$$

When is it negative?

$$R^2_{\text{oos}} < 0 \iff \sum (y_t - \hat{y}_t)^2 > \sum (y_t - \bar{y}_{\text{test}})^2$$

In words: the model's test-set squared error is larger than the squared error from simply predicting the holdout mean.

- Gold daily returns are noisy, so negative holdout R^2 is not unusual in short forecasting windows.
- A negative R^2 does **not** mean the model is useless; it means it fails to beat the holdout-mean baseline under squared-error loss.
- In this project, **RMSE**, **MAE**, and directional accuracy are more informative summary metrics than R^2 alone.

Conclusion

Answer to our research questions

1. We can operationalize Trump's short-horizon pressure-then-soften pattern into an event-level probability signal, *P_taco*.
2. That signal provides incremental information beyond standard macro and market controls.
3. A two-stage structure works better than direct text aggregation for this forecasting task.

These conclusions should be understood under the current definition, sample boundary, and validation protocol.

Appendix: Validation Protocol Backup

- Outer CV determines the final model comparison.
- Inner CV handles model-specific tuning.
- Stage 1 probabilities passed to Stage 2 are always OOF or true holdout predictions.
- Purged CV and embargo reduce nearby-sample contamination.
- Lagged macro / market controls preserve time-availability discipline.

Appendix: Labeling Logic Backup

- Start from candidate pressure-like events.
- For each anchor event, inspect Trump's own follow-up texts within the next 48 hours.
- Use DeepSeek-assisted prompting to judge whether a retreat signal appears.
- The final Stage 1 target is therefore an operational *text-observed retreat* label.